Fractions, Decimals and Percentages

Name _____ Date _____ No calculator Version 1

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KEY WORDS fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · per cent *(ōrau)* · equivalent *(ōrite)*

A Percentages and decimals ●○○

⚠ **Watch out:** Per cent means "out of 100", so divide by 100: 8% is 0.08, not 0.8.

1. Write each percentage as a decimal.

a) 50% = _____ **b)** 25% = _____ **c)** 8% = _____ **d)** 70% = _____

e) 3% = _____ **f)** 120% = _____ **g)** 12.5% = _____ **h)** 0.5% = _____

2. Write each decimal as a percentage.

a) 0.45 = _____ **b)** 0.7 = _____ **c)** 0.09 = _____ **d)** 0.3 = _____

e) 1.5 = _____ **f)** 0.125 = _____ **g)** 0.01 = _____ **h)** 2 = _____

B Fractions to decimals and percentages ●●○

💡 **Remember:** A fraction is a division: $\frac{3}{4} = 3 \div 4 = 0.75$. Then multiply by 100 for the percentage: 75%.

3. Write each fraction as a decimal.

a) $\frac{1}{2}$ = _____ **b)** $\frac{3}{4}$ = _____ **c)** $\frac{2}{5}$ = _____ **d)** $\frac{7}{10}$ = _____

e) $\frac{1}{8}$ = _____ **f)** $\frac{9}{20}$ = _____ **g)** $\frac{3}{25}$ = _____ **h)** $\frac{3}{2}$ = _____

4. Write each fraction as a percentage.

a) $\frac{1}{4}$ = _____ **b)** $\frac{1}{5}$ = _____ **c)** $\frac{3}{10}$ = _____ **d)** $\frac{4}{5}$ = _____

e) $\frac{11}{20}$ = _____ **f)** $\frac{9}{25}$ = _____ **g)** $\frac{7}{50}$ = _____ **h)** $\frac{5}{4}$ = _____

C Writing fractions ●●○

💡 **Remember:** Write the percentage over 100, then simplify: $40\% = \frac{40}{100} = \frac{2}{5}$.

5. Write each percentage as a fraction in its simplest form.

a) 50% = _____ **b)** 25% = _____ **c)** 40% = _____ **d)** 90% = _____

e) 15% = _____ **f)** 64% = _____ **g)** 5% = _____ **h)** 1% = _____

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6. Write each decimal as a fraction in its simplest form.

a) 0.3 = _____ **b)** 0.5 = _____ **c)** 0.25 = _____ **d)** 0.8 = _____

e) 0.45 = _____ **f)** 0.06 = _____ **g)** 0.125 = _____ **h)** 0.72 = _____

D All three together ●●○

7. Complete the table. Write each fraction in its simplest form.

Fraction	Decimal	Percentage
	0.25	
		10%
$\frac{3}{5}$		
	0.45	
		8%
$\frac{3}{8}$		

E Comparing and ordering ●●●

⚠ **Watch out:** Do not compare the digits you can see. Change everything to the same form first (percentages are usually the easiest).

8. Which is larger?

a) $\frac{1}{2}$ or 0.45 **b)** 30% or $\frac{1}{4}$ **c)** 0.7 or 68%

d) $\frac{3}{5}$ or 65% **e)** 0.08 or $\frac{1}{10}$ **f)** $\frac{3}{8}$ or 40%

9. Write each list in order, smallest first.

a) $\frac{1}{2}, \quad 0.4, \quad 45\%$ **b)** $70\%, \quad \frac{3}{4}, \quad 0.72$

c) $8\%, \quad 0.7, \quad \frac{3}{4}, \quad 72\%$ **d)** $\frac{9}{10}, \quad 0.95, \quad 89\%, \quad \frac{4}{5}$

F Problems ●●●

10. Aroha scores 18 out of 25 in a test. Ben scores 70% in the same test. Who did better? Show your working.

11. $\frac{3}{5}$ of the students in a class play a sport. What percentage of the class do not play a sport?

★ Extension

12. Write 0.5% as a fraction in its simplest form.

13. Write 250% as a mixed number in its simplest form.

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Fractions, Decimals and Percentages

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KEY WORDS fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · per cent *(ōrau)* · equivalent *(ōrite)*

A Percentages and decimals ●○○

⚠ **Watch out:** Per cent means "out of 100", so divide by 100: 8% is 0.08, not 0.8.

1. Write each percentage as a decimal.

a) 75% = _____ **b)** 40% = _____ **c)** 6% = _____ **d)** 90% = _____

e) 2% = _____ **f)** 150% = _____ **g)** 37.5% = _____ **h)** 0.2% = _____

2. Write each decimal as a percentage.

a) 0.35 = _____ **b)** 0.9 = _____ **c)** 0.04 = _____ **d)** 0.6 = _____

e) 1.2 = _____ **f)** 0.375 = _____ **g)** 0.07 = _____ **h)** 3 = _____

B Fractions to decimals and percentages ●●○

Remember: A fraction is a division: $\frac{3}{4} = 3 \div 4 = 0.75$. Then multiply by 100 for the percentage: 75%.

3. Write each fraction as a decimal.

a) $\frac{1}{4}$ = _____ **b)** $\frac{1}{5}$ = _____ **c)** $\frac{3}{5}$ = _____ **d)** $\frac{9}{10}$ = _____

e) $\frac{3}{8}$ = _____ **f)** $\frac{7}{20}$ = _____ **g)** $\frac{6}{25}$ = _____ **h)** $\frac{5}{2}$ = _____

4. Write each fraction as a percentage.

a) $\frac{1}{2}$ = _____ **b)** $\frac{3}{4}$ = _____ **c)** $\frac{7}{10}$ = _____ **d)** $\frac{2}{5}$ = _____

e) $\frac{13}{20}$ = _____ **f)** $\frac{8}{25}$ = _____ **g)** $\frac{9}{50}$ = _____ **h)** $\frac{7}{4}$ = _____

C Writing fractions ●●○

Remember: Write the percentage over 100, then simplify: $40\% = \frac{40}{100} = \frac{2}{5}$.

5. Write each percentage as a fraction in its simplest form.

a) 75% = _____ **b)** 20% = _____ **c)** 60% = _____ **d)** 30% = _____

e) 35% = _____ **f)** 48% = _____ **g)** 4% = _____ **h)** 1% = _____

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6. Write each decimal as a fraction in its simplest form.

a) 0.7 = _____ **b)** 0.2 = _____ **c)** 0.75 = _____ **d)** 0.4 = _____

e) 0.35 = _____ **f)** 0.08 = _____ **g)** 0.375 = _____ **h)** 0.64 = _____

D All three together ●●○

7. Complete the table. Write each fraction in its simplest form.

Fraction	Decimal	Percentage
	0.5	
		20%
$\frac{4}{5}$		
	0.35	
		12%
$\frac{5}{8}$		

E Comparing and ordering ●●●

> ⚠ **Watch out:** Do not compare the digits you can see. Change everything to the same form first (percentages are usually the easiest).

8. Which is larger?

a) $\frac{1}{2}$ or 0.55 **b)** 20% or $\frac{1}{4}$ **c)** 0.6 or 58%

d) $\frac{2}{5}$ or 45% **e)** 0.09 or $\frac{1}{10}$ **f)** $\frac{5}{8}$ or 60%

9. Write each list in order, smallest first.

a) $\frac{1}{2}, \quad 0.6, \quad 55\%$ **b)** $80\%, \quad \frac{3}{4}, \quad 0.78$

c) $6\%, \quad 0.5, \quad \frac{11}{20}, \quad 52\%$ **d)** $\frac{7}{10}, \quad 0.75, \quad 69\%, \quad \frac{4}{5}$

F Problems ●●●

10. Aroha scores 21 out of 25 in a test. Ben scores 80% in the same test. Who did better? Show your working.

11. $\frac{3}{4}$ of the students in a class play a sport. What percentage of the class do not play a sport?

★ Extension

12. Write 2.5% as a fraction in its simplest form.

13. Write 350% as a mixed number in its simplest form.

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